

Joe Cameron Blake Wolters

Pre-Medical Researcher — AI Medical Imaging, Clinical Data, and Digital Media

joewolters.github.io

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EDUCATION

Bachelor of Science in Biology, Medical Concentration, University of South Florida, Tampa, FL — Expected May 2028

Cumulative GPA: 3.83

High School Diploma, Southwest Florida Christian Academy, Fort Myers, FL — 2018 to 2024

Graduated *Summa Cum Laude* · Weighted GPA: 4.02 · Unweighted GPA: 3.67

RESEARCH EXPERIENCE

Neurosurgery Research Assistant (Intern)

LSU Health Sciences Center Shreveport, Department of Neurosurgery — May 2025 to July 2025

- Contributed to an AI-driven MRI research project predicting Vertebral Bone Quality (VBQ) scores for preoperative spine surgery decision-making.
- Processed and organized over 1,000 MRI studies; manually reviewed and validated 400+ VBQ scores for publication preparation.
- Performed segmentation and image analysis using 3D Slicer, ITK-SNAP, and MicroDICOM in a structured clinical research workflow.
- Co-authored a manuscript currently under peer review at *Neurosurgery* (see Manuscripts Under Review section).

MANUSCRIPTS UNDER REVIEW / SUBMITTED

- Quinones CJ, Jameson W, Schwartz A, Lupo B, **Wolters J**, Kumbhare D, Guthikonda B, Hoang S. *Automated Quantification of Vertebral Bone Quality Score Using an Artificial Intelligence Framework*. Neurosurgery. Submitted September 2025.

TECHNICAL SKILLS

Medical Imaging & Research: 3D Slicer (segmentation, MRI analysis), ITK-SNAP, MicroDICOM, manual VBQ scoring, clinical dataset validation.

AI-Assisted Development: Daily working knowledge of Anthropic Claude (including Claude Code), OpenAI ChatGPT, and a custom Model Context Protocol (MCP) server (code-review-graph) integrated into a personal production project.

Web & Backend: HTML5, CSS3, vanilla JavaScript (ES modules), Node.js + Express, Firebase (Auth, Firestore, Storage, Hosting), npm, Playwright (end-to-end testing), Git/GitHub, VS Code, DNS / custom-domain configuration.

Audio/Video & Computer Vision: FFmpeg, fluent-ffmpeg, multer; spectral-flux signal processing; YOLOv8 / Ultralytics (Python).

Productivity & Creative: Microsoft Excel, PowerPoint, Word; Google Workspace; CapCut (video editing), Adobe Photoshop.

PROJECTS

Founder, Solo Developer & Maintainer — realanimereviews.com

Personal full-stack web project — 2024 to Present · github.com/joewolters/real-anime-reviews

- Solo-built and shipped a production web application: vanilla HTML/CSS/JS frontend, Firebase (Auth, Firestore, Cloud Storage, Hosting) backend, custom domain via Namecheap DNS. ~5,000 lines of code across responsive desktop and mobile layouts.
- Implemented authentication (email/password, Google sign-in, password reset), favorites and watchlists, an avatar upload pipeline (client-side image downscaling), real-time comments and voting, community reviews with discussion threads, and a notifications inbox.
- Designed a five-collection-family Firestore data model with composite indexes; used onSnapshot listeners throughout for real-time updates without page refresh.
- Stood up an end-to-end Playwright test suite (7 specs) and ran a self-led code/security audit that fixed real production deploy-hygiene issues.
- Built largely with Anthropic Claude Code and OpenAI ChatGPT; integrated a custom code-review-graph MCP server for graph-based code review; project-level CLAUDE.md codifies AI-collaboration rules.

Founder & Solo Developer — [PickleClipper](#) (work in progress)

Personal Node.js + FFmpeg + Python project — 2025 to Present · github.com/joewolters/PickleClipper

- Solo-building a local web application that automatically trims raw pickleball footage to the rally action; eventual scope includes player and ball tracking, in-or-out call analysis, and habit analytics.
- Implemented audio-detection layer in vanilla JavaScript (spectral-flux onset detection, peak-shape filtering, custom solo-peak rally clustering). Approximately 92% recall across three labeled outdoor singles test clips (F1 0.768–0.881).
- Express server with multer (8 GB cap for iPhone HEVC source), an FFmpeg trim/stitch pipeline, and a test harness that emits per-clip F1, precision, and recall metrics; parameter-autotune harness for sweeping algorithm settings.
- Currently transitioning to the visual-analysis phase: integrating Ultralytics YOLOv8 (Python) for player tracking. Project-level CLAUDE.md codifies the bounded-autonomy AI-collaboration rules used throughout development.

Founder, Video Editor & Content Creator — [lowlvledits](#)

[YouTube](#) · [Instagram](#) (@lowlvledits) — Active

- Produces and edits short- and long-form video content under the @lowlvledits brand.

- Owns the full production pipeline: footage selection, editing in CapCut, color and audio adjustments, captioning, thumbnails, posting, and audience engagement.

CLINICAL EXPERIENCE

Medical Shadowing — Emergency Medicine

Archbold Hospital, Thomasville, GA

- Shadowed emergency room physicians during patient assessments, critical decision-making, and emergency procedures.
- Observed triage, patient stabilization, and multidisciplinary collaboration in a high-acuity hospital environment.
- Developed familiarity with physician–patient communication, clinical professionalism, and ER workflow.

LEADERSHIP & ACTIVITIES

Risk Manager, Executive Board — USF Running Club

University of South Florida, Tampa, FL — Spring 2025 to Present

- Elected to the Executive Board to oversee safety, risk management, and emergency response for all club activities.
- Maintain active CPR certification per the USF Running Club Constitution.
- Coordinate with USF administration on event compliance, emergency planning, and travel logistics.
- Lead fundraising for the club's 501(c)(3) account, including partner-restaurant outreach and event coordination.

Captain — Cross-Country and Track & Field

Southwest Florida Christian Academy, Fort Myers, FL — 2020 to 2024 (4 consecutive years)

- Served as team captain for both Cross-Country and Track & Field across all four years of high school competition.
- Led practices, set the tone for team culture, and mentored underclassmen on training, racing strategy, and competition mindset.

WORK EXPERIENCE

Event Server — Good Food Catering Company

Tampa, FL — September 2024 to Present

- Provide professional food and beverage service at private and corporate events; coordinate setup, service, and breakdown to ensure smooth, high-quality guest experiences.

Warehouse Assistant — Sensible Spa Solutions

Fort Myers, FL — January 2019 to 2023

- Managed inventory organization and product restocking; filed warranty service claims and coordinated with vendors and customers.

VOLUNTEER & COMMUNITY SERVICE

- **USF Bull Haul Move-In Volunteer**, University of South Florida, 2024 to 2025.
- **Hurricane Cleanup Volunteer**, Fort Myers, FL, 2022 — cleared debris, distributed essential supplies, and assisted affected communities in disaster relief efforts.
- **Track & Cross-Country Event Volunteer**, Fort Myers, FL, 2019 to 2022.

HONORS & AWARDS

- Co-Author on submitted manuscript — *Neurosurgery*, 2025
- ASI Distinguished Student Award — Academics, Leadership, & Fine Arts
- Graduated *Summa Cum Laude*, Southwest Florida Christian Academy, 2024
- National Honor Society Member — Grades 10, 11, 12
- Cross-Country & Track Captain — 4 consecutive years (high school)

LANGUAGES

- English (native)